Kalan Brunell

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Education

Tufts University, Medford, MA — Expected May 2027
Bachelor of Science in Computer Engineering — *Dean's List*

Experience

Avionics Cybersecurity Intern, MITRE, RCAT Lab - Bedford, MA — May 2026 - Present

- Developing SystemVerilog FPGA IP for a data bus processing device that enforces modern cybersecurity principles on legacy and emerging avionics buses.
- Researching and testing hardware-driven data processing architectures to improve throughput and decrease latency for real-time avionics applications.
- Implemented FPGA USB and SPI interfaces for configuration, control, and host data exchange.
- Designing, simulating, and verifying RTL with CocoTB and Vivado on AMD Xilinx FPGAs.

Manufacturing Automation Engineering Intern, Gentex Corporation - Manchester, NH — Summer 2025

- Designed a custom 2D inspection system to increase manufacturing inspection accuracy by up to 60%.
- Developed a custom microcontroller-based protocol converter to interface the vision system with manufacturing robotic arms to enable real-time object location and automated quality inspections.
- Developed Fanuc robot programs using the vision system to automate manufacturing tasks like soldering, adhesive application, and part manipulation.

Projects

FSAE Electric Racecar Accumulator, Tufts Electric Racing - Project Manager — May 2025 - Present

- Leading the design of a removable 440-cell, 450V accumulator for a Formula SAE Electric racecar.
- Designing a custom STM32 board for cell monitoring, balancing, charge control, and CAN telemetry with notably improved size, weight, and power efficiency over OTS solutions.
- Engineering a custom RS-485 protocol for real-time temperature mapping and fault reporting across 4 distributed thermistor boards.

MITRE Embedded Capture the Flag (eCTF), Tufts University — January 2026 - April 2026

- Designed and implemented a secure embedded device to meet challenge requirements, then analyzed and exploited competing teams' designs.
- Implemented secure firmware features on an ARM Cortex microcontroller to mitigate hardware-based vulnerabilities such as buffer overflows and side-channel attack vectors.
- Probed attack surfaces including UART interfaces, flash memory access, and fault injection vectors to uncover security weaknesses.

5-Stage Pipelined ARM Processor, Tufts University — September 2025 - December 2025

- Engineered a 5-stage pipelined processor with hazard detection and forwarding in VHDL.
- Features IF/ID/EX/MEM/WB stages with support for arithmetic, logic, memory, and branch instructions.
- Implemented hazard detection and forwarding units to resolve data and control hazards, optimizing instruction throughput.

Skills

Programming: C, C++, Python, ARM Assembly, RISC, CMSIS, OOP, Scripting
FPGA & Digital Design: SystemVerilog, VHDL, RTL Design, CocoTB, GTKWave, Vivado, Lattice Radiant
Embedded & Electronics: STM32, Bare-Metal & RTOS Firmware, Peripheral Interfacing, I2C/SPI/UART/CAN/RS-485, ADCs, GDB
Circuit & PCB Design: Altium Designer, KiCad, LTspice, Electrical Lab Equipment, Soldering
Other Tools: Microsoft Office, MATLAB, SolidWorks, GIT, LaTeX